

Irish Math Olympiad 1989 — P4/10

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Note that $12^2 = 144$ ends in two 4's and $38^2 = 1444$ end in three 4's. Determine the length of the longest string of equal nonzero digits in which the square of an integer can end.

Solution

As hinted at in the problem statement, the answer is exactly 3.

Claim 1: If a perfect square ends with two or more equal digits, then the last digit of that number must be 4.

Proof: Let x^2 be the perfect square which ends in two or more equal digits. Let the last digit of x^2 be d . Perfect squares can only be $\equiv 0, 1$ modulo 4.

- If $x^2 \equiv 0 \pmod{4}$ then the number $\overline{dd}$ is divisible by 4, due to the divisibility rule for 4. Checking each case we find that d can only be 0, 4 or 8, but since the problem wants "equal non-zero digits" d can only be 4 or 8.
- If $x^2 \equiv 1 \pmod{4}$ then the number $\overline{d(d-1)}$ is divisible by 4, due to the divisibility rule for 4. Again by checking each case, d can only be 3 or 7.

Now if one lists out all the perfect squares mod 10, you find that x^2 can only be congruent to 0, 1, 4, 5, 6, 9 modulo 10. This filters our options down to only $d = 4$, as desired. $\square$

Claim 2: If a perfect square x^2 ends with two or more equal digits, then the units digit of x can only be 2 or 8.

Proof: Let u be the units digit of x . Note that $x - u \equiv 0 \pmod{10}$ because $x - u$ is just the number x but with a units digit of 0.

Now since $x^2 = ((x - u) + u)^2 = (x - u)^2 + 2u(x - u) + u^2 \equiv u^2 \pmod{10}$, we need u^2 to end in a 4, due to claim 1. Now by checking each case we determine that u can only possibly be 2 or 8. $\square$

Claim 3: If a perfect square x^2 ends with two or more equal digits, then x ends with the digits $\overline{38}$ or $\overline{12}$, provided that x has at least 2 digits in it's decimal representation.

Proof: Let u be the units digit of x . Let t be the tens digit of x . Note that $x - 10t - u \equiv 0 \pmod{100}$, because $x - 10t - u$ is just the number x with units digit 0 and tens digit 0 using the fact that $x \geq 100$.

Now $x^2 = ((x - 10t - u) + (10t + u))^2 = (x - 10t - u)^2 + 2(10t + u)(x - 10t - u) + (10t + u)^2 \equiv (10t + u)^2 \pmod{100}$ shows us that x^2 ends in $(10t + u)^2 \pmod{100}$. We need $(10t + u)^2 \equiv 20tu + u^2 \equiv 44 \pmod{100}$ due to claim 1. We need only to check the cases $u = 2, 8$ due to claim 2.

- If $u = 2$ then $40t + 4 \equiv 44 \pmod{100}$, and so $40(t - 1) \equiv 0 \pmod{100}$ which gives $t = 1$. So x ends in $\overline{12}$.

- If $u = 8$ then $60t + 64 \equiv 44 \pmod{100}$, and so $60t \equiv 80 \pmod{100}$ which gives $t = 3$. So x ends in $\overline{38}$.

□

Now finally if x has at least three digits and ends in three consecutive digits then, let h be the hundreds digit of x , let t be the tens digit of x , let u be the units digit of x .

Then since $x^2 = ((x - 100h - 10t - u) + (100h + 10t + u))^2 \equiv (100h + 10t + u)^2 \pmod{1000}$, we need $t^2 + u^2 + 200th + 200uh + tu \equiv 444 \pmod{1000}$.

If x ends in $\overline{38}$ then we have $9 + 64 + 1600h + 24 \equiv 444 \pmod{1000}$ and so $600h \equiv 347 \pmod{1000}$ which can be checked to have no solutions.

Similarly if x ends in $\overline{12}$ then we have $1 + 4 + 200h + 400h + 2 \equiv 444 \pmod{1000}$ and so $600h \equiv 437 \pmod{1000}$ which can again be checked to have no solutions.

So x has at most 2 digits and so the answer turns out to be $x = 38$ with $38^2 = 1444$ ending in three 4's. □